

# Enterprise Network Design using Cisco Packet Tracer

## Project Summary

Designed and implemented an enterprise network using Cisco Packet Tracer featuring VLAN segmentation, Router-on-a-Stick, Inter-VLAN Routing, DHCP, DHCP Relay, DNS, and automatic IP address allocation. The project demonstrates how a small enterprise network can be organized, secured, and managed efficiently.

## Project Information

| Field                  | Value                          |
|------------------------|--------------------------------|
| <b>Project Name</b>    | Enterprise Network Design      |
| <b>Platform</b>        | Cisco Packet Tracer 9.0.0.0810 |
| <b>Author</b>          | Bala Venkata Sai Dindu         |
| <b>Domain</b>          | Computer Networking            |
| <b>Project Type</b>    | Enterprise Network Simulation  |
| <b>Completion Date</b> | July 2026                      |

## Project Objectives

The primary objective of this project was to design and implement a scalable enterprise network using Cisco Packet Tracer while applying industry-standard networking concepts.

The project focuses on:

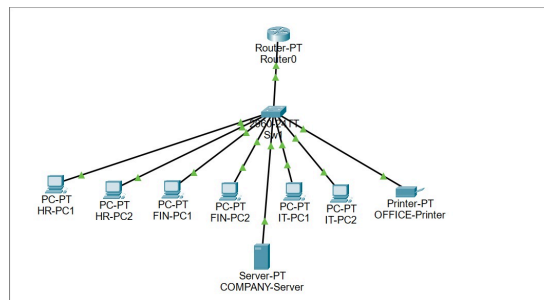
- Designing a logically segmented enterprise network using VLANs.
- Reducing broadcast traffic by creating multiple broadcast domains.
- Implementing Router-on-a-Stick for Inter-VLAN communication.
- Automating IP address allocation using DHCP.
- Configuring DHCP Relay ( `ip helper-address` ) to serve clients across VLANs.

- Validating end-to-end connectivity through network testing.

## Technologies Used

| Technology          | Purpose                      |
|---------------------|------------------------------|
| Cisco Packet Tracer | Network Simulation           |
| VLAN                | Logical Network Segmentation |
| Access Ports        | Connect End Devices          |
| Trunk Ports         | Carry Multiple VLANs         |
| IEEE 802.1Q         | VLAN Tagging                 |
| Router-on-a-Stick   | Inter-VLAN Routing           |
| DHCP                | Automatic IP Assignment      |
| DHCP Relay          | Forward DHCP Requests        |
| DNS                 | Name Resolution              |
| ICMP                | Connectivity Testing         |

## Enterprise Network Topology



This topology represents a small enterprise network consisting of three departments:

- Human Resources
- Finance
- IT

Each department is placed in its own VLAN to improve network organization and security.

The switch connects to the router through a trunk link, enabling Router-on-a-Stick for communication between VLANs

## VLAN Design

```
Switch>show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                           Gig0/2
10   HR                     active    Fa0/4, Fa0/2, Fa0/7
20   FINANCE                active    Fa0/3, Fa0/4
30   IT                     active    Fa0/5, Fa0/6, Fa0/8
1002 fddi-default          active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active
Switch>
```

## VLAN Allocation

| VLAN ID | Department | Network         |
|---------|------------|-----------------|
| 10      | HR         | 192.168.10.0/24 |
| 20      | Finance    | 192.168.20.0/24 |
| 30      | IT         | 192.168.30.0/24 |

## Trunk Port Configuration

```
Switch>show interface trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/24    on        802.1q         trunking      1
Port      Vlans allowed on trunk
Fa0/24    1-1008
Port      Vlans allowed and active in management domain
Fa0/24    1,10,20,30
Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,10,20,30
Switch>
```

The connection between the switch and the router is configured as a trunk.

This allows frames from multiple VLANs to travel through a single Ethernet cable.

The switch inserts an IEEE 802.1Q VLAN tag before forwarding frames over the trunk and removes the tag before delivering the frame to the destination end device.

## Router-on-a-Stick

```
Router#show ip interface brief
Interface    IP-Address      OK? Method Status  Protocol
FastEthernet0/0  unassigned      YES manual up       up
FastEthernet0/10 192.168.10.1    YES manual up       up
FastEthernet0/20 192.168.20.1    YES manual up       up
FastEthernet0/30 192.168.30.1    YES manual up       up
FastEthernet1/0  unassigned      YES unset administratively down down
Serial2/0       unassigned      YES unset administratively down down
Serial3/0       unassigned      YES unset administratively down down
FastEthernet4/0  unassigned      YES unset administratively down down
FastEthernet5/0  unassigned      YES unset administratively down down
Router#
```

The router uses Router-on-a-Stick by creating one subinterface for each VLAN

| Interface | VLAN    | Gateway      |
|-----------|---------|--------------|
| Fa0/0.10  | VLAN 10 | 192.168.10.1 |
| Fa0/0.20  | VLAN 20 | 192.168.20.1 |
| Fa0/0.30  | VLAN 30 | 192.168.30.1 |

## Packet Flow

PC → Switch

↓

Switch adds VLAN Tag

↓

Trunk Port

↓

Router

↓

Destination VLAN identified

↓

Packet Routed

↓

Switch removes VLAN Tag

↓

Destination PC

## DHCP Configuration

|                                                 |        |                 |         |             |            |
|-------------------------------------------------|--------|-----------------|---------|-------------|------------|
| Physical                                        | Config | <u>Services</u> | Desktop | Programming | Attributes |
| <b>SERVICES</b>                                 |        |                 |         |             |            |
| HTTP                                            |        |                 |         |             |            |
| SMTP                                            |        |                 |         |             |            |
| DHCPv6                                          |        |                 |         |             |            |
| TFTP                                            |        |                 |         |             |            |
| DNS                                             |        |                 |         |             |            |
| STUN/SDP                                        |        |                 |         |             |            |
| ASA                                             |        |                 |         |             |            |
| NTP                                             |        |                 |         |             |            |
| SNMP                                            |        |                 |         |             |            |
| FTP                                             |        |                 |         |             |            |
| SSH                                             |        |                 |         |             |            |
| Voice Management                                |        |                 |         |             |            |
| Radius EAP                                      |        |                 |         |             |            |
| PPPoE                                           |        |                 |         |             |            |
| Interface: (FastEthernet0/24) Service: On / Off |        |                 |         |             |            |
| Pool Name: pool1                                |        |                 |         |             |            |
| Default Gateway: 192.168.10.1                   |        |                 |         |             |            |
| DNS Server: 192.168.30.100                      |        |                 |         |             |            |
| Start IP Address: 192                           |        |                 |         |             |            |
| End IP Address: 199                             |        |                 |         |             |            |
| Subnet Mask: 255                                |        |                 |         |             |            |
| Maximum Number of Users: 100                    |        |                 |         |             |            |
| TFTP Server: 0.0.0.0                            |        |                 |         |             |            |
| VLC Address: 0.0.0.0                            |        |                 |         |             |            |
|                                                 |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| Start IP Address                                |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
| DNS Server                                      |        |                 |         |             |            |
| Default Gateway                                 |        |                 |         |             |            |
| Pool Name                                       |        |                 |         |             |            |
| Add                                             |        |                 |         |             |            |
| Save                                            |        |                 |         |             |            |
| Remove                                          |        |                 |         |             |            |
| VLC Management                                  |        |                 |         |             |            |
| Radius                                          |        |                 |         |             |            |
| TFTP Server                                     |        |                 |         |             |            |
| Max User                                        |        |                 |         |             |            |
| Subnet Mask                                     |        |                 |         |             |            |
|                                                 |        |                 |         |             |            |

Three DHCP pools were configured.

| Pool    | Network         |
|---------|-----------------|
| HR      | 192.168.10.0/24 |
| Finance | 192.168.20.0/24 |
| IT      | 192.168.30.0/24 |

The router was configured with `ip helper-address` on VLAN 10 and VLAN 20 subinterfaces to forward DHCP requests to the DHCP server located in VLAN 30.

## Testing and Validation

```
Command Prompt
C:\>ping 192.168.30.10
Pinging 192.168.30.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 1ms
C:\>ping 192.168.20.100
Pinging 192.168.20.100 with 32 bytes of data:
Reply from 192.168.20.100: bytes=32 time=1ms TTL=127
Reply from 192.168.20.100: bytes=32 time=1ms TTL=127
Reply from 192.168.20.100: bytes=32 time=1ms TTL=127
Reply from 192.168.20.100: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 1ms
C:\>
```

## Connectivity Tests

| Test                 | Status    |
|----------------------|-----------|
| HR → HR              | ✓ Success |
| HR → Finance         | ✓ Success |
| HR → IT              | ✓ Success |
| DHCP Assignment      | ✓ Success |
| DNS Configuration    | ✓ Success |
| Printer Connectivity | ✓ Success |
| Server Connectivity  | ✓ Success |

## Configuration Files

Router\_Configuration.txt

Switch\_Configuration.txt

## Key Networking Concepts Demonstrated

- VLAN Segmentation
- Access Ports
- Trunk Ports
- IEEE 802.1Q VLAN Tagging
- Router-on-a-Stick
- Inter-VLAN Routing
- DHCP
- DHCP Relay ( `ip helper-address` )
- DNS
- ICMP Connectivity Testing

## Skills Demonstrated

- Enterprise Network Design
- VLAN Planning
- Router Configuration
- Switch Configuration
- Network Troubleshooting
- DHCP Configuration
- Inter-VLAN Routing
- Documentation
- Cisco Packet Tracer

## Conclusion

This project successfully demonstrates the design and implementation of a small enterprise network using Cisco Packet Tracer. By combining VLAN segmentation,

Router-on-a-Stick, Inter-VLAN Routing, DHCP, and DHCP Relay, the network provides organized, scalable, and manageable communication between departments. The troubleshooting process also reinforced the importance of systematic diagnosis and validation during network deployment.